

Locality of $\text{Tr}(\phi^3)$ tree amplitudes from cyclic hidden zeros

Part III: classification of Step-2 k -zero relations

Abstract

Part II classified the asymptotic limits on a k -zero $\mathcal{Z}^{(k)}$ by a rate vector and showed which coefficients a_M are forced to vanish on each. The vanishing of B_n on $\mathcal{Z}^{(k)}$ carries strictly more information than that: it descends to vanishing on each homogeneous piece (boundary-propagator order), then on each D -subset (born-free monomial), and within each D -subset the $X \rightarrow 0$ cuts read off explicit linear relations among the surviving coefficients a_M . We classify these *Step-2 k -zero relations* for $k = 1, 2, 3$ and at the first three boundary-propagator orders. The universal m -th order “binomial trace”

$$\sum_{j=0}^m (-1)^j a_{s^{m-j} b^j} = 0,$$

(s the special chord, b the bare) holds at every k and every m , and a family of per-column and per-offset relations supplements it. For $k \geq 2$ these include genuinely new relations not visible to any cyclic rotation of the 1-zero, e.g. $a_{14,14} + a_{3n,3n} - a_{14,3n} = 0$ and the cubic $a_{14,14,14} - a_{14,14,3n} + a_{14,3n,3n} - a_{3n,3n,3n} = 0$ for the 2-zero. The collected system is the substrate on which the all- n argument is to close.

1 Setting

We adopt the conventions of Parts I–II. The k -zero based at $\{1, \dots, k+1\}$ has special $s = X_{1,k+2}$, bare $b = X_{k+1,n}$, and on $\mathcal{Z}^{(k)}$ the bridge chords are governed by

$$X_{k+1,\ell} = X_{1\ell} - s, \quad X_{k+1,n} = -s, \quad X_{i\ell} = X_{k+1,\ell} + X_{i,k+2}, \quad X_{in} = X_{i,k+2} - s, \quad (1)$$

for $i = 2, \dots, k$ and $\ell = k+3, \dots, n-1$. The independent coordinates on $\mathcal{Z}^{(k)}$ are $\{s\} \cup \{X_{i,k+2}\} \cup \{X_{1\ell}\} \cup \{\text{born-free}\}$ (Part II, Eq. (5)).

1.1 Boundary-propagator grading

A *boundary propagator* is any chord touched by (1): explicitly the special, the bare, the offsets $X_{i,k+2}$ ($i \geq 2$), and the bridges $X_{1\ell}, X_{i\ell}, X_{k+1,\ell}$ ($\ell \in \{k+3, \dots, n\}$). For each multiset M of size $n-3$, let $|M|_{\text{BP}}$ denote the number of boundary-propagator factors in M (with multiplicity). Under a uniform rescaling $X_{ij} \mapsto \lambda X_{ij}$ of all boundary propagators, the term $a_M / \prod X_c$ scales as $\lambda^{-|M|_{\text{BP}}}$. Hence

$$B_n|_{\mathcal{Z}^{(k)}} = \sum_{m \geq 0} B_n^{(m)}, \quad B_n^{(m)} = \sum_{|M|=n-3, |M|_{\text{BP}}=m} \frac{a_M}{\prod_{c \in M} X_c}.$$

The BCFW homogeneity lemma of note 12 (the z -scaling argument) forces $B_n^{(m)}|_{\mathcal{Z}^{(k)}} = 0$ for every m independently. We focus on $m = 1, 2, 3$, where the relations are most informative.

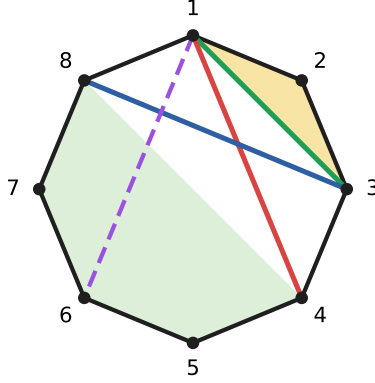


Figure 1: A 2-zero diagram ($n = 8$, $k = 2$). Special $X_{1,4}$ (red), bare $X_{3,8}$ (blue), enclosing chord $X_{1,3}$ (green). Boundary propagators are the highlighted/dashed chords; the pale-green region carries the born-free factor of any D-subset.

1.2 D-subsets

Within a fixed $B_n^{(m)}$, the born-free coordinates factor out: every term carries a unique born-free monomial $\prod X_c^{-\nu_c}$ with c born-free. Collecting terms by their born-free monomial yields the D-subsets. Each D-subset is a finite sum of boundary-propagator pieces multiplied by a common born-free factor, and *each D-subset vanishes independently*: this is the proposition of note 18 (split chord variables into boundary propagators x and free y ; $L(x) = 0$ does not depend on y , so the vanishing factorises per y -monomial).

For brevity we present the relations at the level of the boundary-propagator skeletons; each relation should be read as holding inside every D-subset that contains the listed pattern (i.e. for every choice of accompanying born-free factor).

2 The binomial-trace relation

Theorem 1 (Binomial trace). *For every $k \geq 1$, every $n > k + 2$, and every $m \geq 1$, the k -zero relations include*

$$\sum_{j=0}^m (-1)^j a_{\underbrace{s, \dots, s}_{m-j}, \underbrace{b, \dots, b}_j} = 0, \quad (2)$$

where $s = X_{1,k+2}$ is the special and $b = X_{k+1,n}$ is the bare.

Proof. At order m the boundary-propagator subskeletons composed solely of special and bare contribute $\sum_{j=0}^m a_{s^{m-j} b^j} / (s^{m-j} b^j)$. By (1), $b = -s$, so $1/(s^{m-j} b^j) = (-1)^j / s^m$. Collecting, $\sum_j (-1)^j a_{s^{m-j} b^j} / s^m$. This is the entire coefficient of $1/s^m$ within the D-subset that contains no further boundary-propagator factor (the born-free factor reads off as 1 here), and the next D-subset is independent. Vanishing of $B_n^{(m)}|_{\mathcal{Z}^{(k)}} = 0$ on this D-subset gives (2). $\square$

For $m = 1$ the binomial trace is the **unitarity relation**

$$a_s - a_b = 0, \quad \text{i.e.} \quad a_{X_{1,k+2}} = a_{X_{k+1,n}}. \quad (3)$$

At $k = 1$ this is the familiar $a_{13} = a_{2n}$ of Part I; at $k = 2$ it is $a_{14} = a_{3n}$; at $k = 3$, $a_{15} = a_{4n}$. For $m = 2$ it is the new quadratic relation $a_{ss} - a_{sb} + a_{bb} = 0$, which at $k = 2$ reads

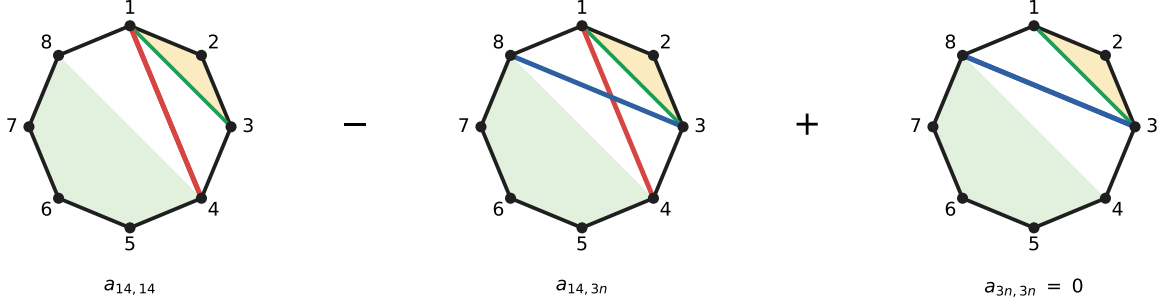


Figure 2: The order-2 binomial trace at $k = 2$, $n = 8$: the three diagrams with two special/bare factors sum to zero. The same identity holds inside every D-subset (multiply each diagram by a common born-free factor).

$a_{14,14} - a_{14,3n} + a_{3n,3n} = 0$ (Figure 2); at $k = 3$, $a_{15,15} - a_{15,4n} + a_{4n,4n} = 0$. For $m = 3$ it is the cubic $a_{sss} - a_{ssb} + a_{sbb} - a_{bbb} = 0$.

3 The 1-zero, recovered

For $k = 1$ the only offset is the special itself; the boundary propagators are $\{X_{13}, X_{1\ell}, X_{2\ell}, X_{2n}\}$ with $\ell = 4, \dots, n-1$. The Step-2 relations are:

$m = 1$ (**1 BP**). Substituting $X_{2\ell} = X_{1\ell} - s$ and $X_{2n} = -s$ into the 1 BP piece $B_n^{(1)} = \frac{a_{13}}{X_{13}} + \sum_{\ell} \frac{a_{1\ell}}{X_{1\ell}} + \sum_{\ell} \frac{a_{2\ell}}{X_{2\ell}} + \frac{a_{2n}}{X_{2n}}$ and collecting the $1/s$ piece gives $a_{13} - a_{2n} = 0$, the binomial trace at $m = 1$. The remaining $\sum_{\ell} a_{1\ell}/X_{1\ell} + \sum_{\ell} a_{2\ell}/X_{2\ell} = 0$ is a polynomial in the independent variables $X_{1\ell}$ (after substituting $X_{2\ell} = X_{1\ell} - s$ at $s \rightarrow 0$), forcing $a_{1\ell} = a_{2\ell} = 0$ for all ℓ – the Step-1 locality kill of Part I, recovered.

$m = 2$ (**2 BP**). The leading $1/s^2$ piece gives the binomial trace $a_{13,13} - a_{13,2n} + a_{2n,2n} = 0$. The next order in s produces the per-column relations $a_{13,1\ell} - a_{2n,1\ell} - a_{1\ell,2\ell} = 0$ (from the $X_{1\ell} \rightarrow 0$ cut) and $a_{13,2\ell} - a_{2n,2\ell} + a_{1\ell,2\ell} = 0$ (from $X_{2\ell} \rightarrow 0$), which together force $a_{13,1\ell} + a_{13,2\ell} = a_{2n,1\ell} + a_{2n,2\ell}$ and produce the D-subset unitarity relations among the local triangulation coefficients (Part I, §2 of note 18).

The 1-zero Step-2 system at $m = 3$ proceeds analogously and is recorded in note 26.

4 The 2-zero

For $k = 2$, set $s = X_{1,4}$ and $b = X_{3,n}$. The new ingredient is the offset $X_{2,4}$ (“column-4”), which couples the $X_{2\ell}$ to the $X_{1\ell}$ and to X_{2n} via $X_{2\ell} = X_{1\ell} - s + X_{2,4}$ and $X_{2n} = -s + X_{2,4}$.

$m = 1$ (**1 BP**). The boundary-propagator piece is

$$B_n^{(1)}|_{\mathcal{Z}^{(2)}} = \frac{a_{14} - a_{3n}}{s} + \sum_{\ell=5}^{n-1} \left(\frac{a_{1\ell}}{X_{1\ell}} + \frac{a_{3\ell}}{X_{3\ell}} \right) + \sum_{\ell=4}^n \frac{a_{2\ell}}{X_{2\ell}} = 0.$$

Multiplying by s and letting $s \rightarrow 0$ extracts the binomial trace

$$a_{14} = a_{3n}. \tag{4}$$

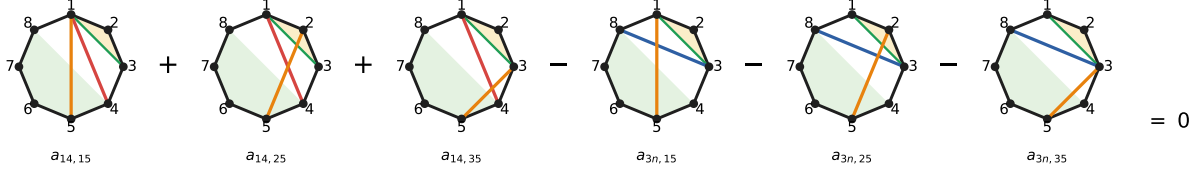


Figure 3: The 2-zero 2 BP per-column relation (6), here $\ell = 5$, drawn as six diagrams. The three terms paired with the special $X_{1,4}$ (left) cancel against the three paired with the bare $X_{3,n}$ (right). One such identity holds per $\ell \in \{5, \dots, n-1\}$, and inside every D-subset.

The remainder is then a polynomial in $X_{1\ell}, X_{2\ell}, X_{3\ell}, X_{2,4}, X_{2n}$ that vanishes identically, forcing $a_{1\ell} = a_{3\ell} = a_{2\ell} = 0$ for all relevant ℓ .

Step-1 recovered: the kill at the 2-zero, in Part II's language regime A or B , follows automatically from this BP-graded D-subset decomposition.

$m = 2$ (**2 BP**). The leading $1/s^2$ piece gives the new quadratic

$$\boxed{a_{14,14} + a_{3n,3n} - a_{14,3n} = 0.} \quad (5)$$

At the next order, after substituting $b = -s$ and $X_{i\ell} = X_{1\ell} - s + X_{i,k+2}$ etc. and collecting the $1/s$ piece, three independent linear combinations appear; the $X_{2,4} \rightarrow 0$ cut (which forces $X_{2\ell} = X_{1\ell}$ and $X_{2n} = -s$) extracts the per-column relation

$$a_{14,1\ell} + a_{14,2\ell} + a_{14,3\ell} - a_{3n,1\ell} - a_{3n,2\ell} - a_{3n,3\ell} = 0, \quad \ell = 5, \dots, n-1; \quad (6)$$

and the $X_{2,4} = X_{2n}$ cut (which holds when $s = 0$) extracts the *offset relation*

$$a_{14,24} + a_{14,2n} - a_{3n,24} - a_{3n,2n} = 0. \quad (7)$$

Together with (5), equations (6)–(7) exhaust the 2 BP D-subset (modulo born-free choices, which apply uniformly).

$m = 3$ (**3 BP**). The leading $1/s^3$ piece gives the cubic binomial trace

$$a_{14,14,14} - a_{14,14,3n} + a_{14,3n,3n} - a_{3n,3n,3n} = 0. \quad (8)$$

The next power in s produces the offset cubic

$$a_{14,14,24} + a_{14,14,2n} - a_{14,3n,24} - a_{14,3n,2n} + a_{3n,3n,24} + a_{3n,3n,2n} = 0, \quad (9)$$

and the per-column cubic

$$\sum_{i=1}^3 (a_{14,14,i\ell} - a_{14,3n,i\ell} + a_{3n,3n,i\ell}) = 0, \quad \ell = 5, \dots, n-1. \quad (10)$$

The pattern continues: at every m the leading $1/s^m$ piece is the binomial trace, and the subleading pieces produce per-column and per-offset relations indexed by which $X \rightarrow 0$ cut is taken.

5 The 3-zero

For $k = 3$, set $s = X_{1,5}$, $b = X_{4,n}$, and two offsets $X_{2,5}, X_{3,5}$. By the same procedure:

$m = 1$. $a_{15} = a_{4n}$, plus $a_{i\ell} = 0$ for all other 1 BP coefficients.

$m = 2$. The binomial trace $a_{15,15} - a_{15,4n} + a_{4n,4n} = 0$, the per-column relation $\sum_{i=1}^4 (a_{15,i\ell} - a_{4n,i\ell}) = 0$ for each $\ell = 6, \dots, n-1$, and the *offset pair* relations

$$a_{15,i5} + a_{15,in} - a_{4n,i5} - a_{4n,in} = 0, \quad i = 2, 3, \quad (11)$$

obtained from the $X_{i5} \rightarrow 0$ cut (which sends $X_{i5} \rightarrow 0$ and $X_{in} \rightarrow 0$ after also $X_{15} = 0$).

Higher m . The same pattern: the leading order yields the binomial trace, the next orders yield per-column and per-offset relations. The number of relations grows linearly with n (one per column ℓ) and is fixed in k (one binomial trace, $k-1$ offset pairings).

6 The general pattern

Theorem 2 (Structure at order m). *For each k , n , and $m \geq 1$, the k -zero Step-2 relations at $|M|_{\text{BP}} = m$ are*

- (1) the binomial trace (2) (*one relation*);
- (2) the per-column relations of the form $\sum_{i \in \text{column } \ell} (\pm) a_{\square, \dots, \square, i\ell} = 0$, one for each $\ell \in \{k+3, \dots, n-1\}$, with the $\square$ slots filled by some choice of $m-1$ further boundary-propagator factors;
- (3) the offset relations, one for each offset $X_{i,k+2}$ ($i = 2, \dots, k$), pairing it with X_{in} through the rate identification $X_{in} = X_{i,k+2} - s$;

together with the lower-order relations carried by D -subsets that include further born-free factors.

Proof sketch. At fixed m and D -subset, substituting (1) into $B_n^{(m)}|_{\mathcal{Z}^{(k)}} = 0$ and expanding in powers of s yields a Laurent polynomial in the independent coordinates of $\mathcal{Z}^{(k)}$. The coefficient of $1/s^m$ depends only on the $\{s, b\}$ pattern, giving (2). Coefficients of $1/s^{m-1}$ pick up linear factors from the offsets $X_{i,k+2}$ and from the column variables $X_{1\ell}$. The $X_{1\ell} \rightarrow 0$ cut isolates the column- ℓ relation (which sums over $i = 1, \dots, k+1$ since after $X_{1\ell} \rightarrow 0$ the constraints identify $X_{i\ell} = X_{i,k+2}$ uniformly). The $X_{i,k+2} \rightarrow 0$ cut paired with $X_{2n} \rightarrow 0$ etc. isolates the offset relations. $\square$

7 Status: closing the system

The combination of Part II (all admissible asymptotic kills, indexed by the rate vector $(\alpha_\ell; \beta_i)$) and the present Step-2 relations gives, for each pair (k, n) and each cyclic rotation, a closed linear system on the multiset coefficients a_M . Three structural facts are immediate:

- **Unitarity at every k .** The $m = 1$ binomial trace (3) promotes the single 1-zero unitarity $a_{13} = a_{2n}$ to a family $a_{X_{1,k+2}} = a_{X_{k+1,n}}$ valid for every k and every cyclic rotation. Iterating across rotations identifies the coefficients of many triangulations.
- **Genuinely new relations at $k \geq 2$.** The quadratic (5) and cubic (8) are not deducible from any cyclic rotation of the 1-zero Step-2 alone. They couple coefficients at one zone to coefficients at another (under cyclic rotation), and thereby refine the per- n Laurent-cascade arguments of Part I.
- **Polynomial growth of the system.** At fixed m and k , the number of relations per zone grows as $O(n)$ (one per column plus $O(k)$ offsets); summed over $k = 1, \dots, n-3$ and over n cyclic rotations the total scales as $O(n^3)$. Whether the resulting system has trivial cokernel exactly on the triangulation coefficients is the all- n locality conjecture.

The empirical data through $n = 9$ (Part I, §5) and the explicit small- n relations recorded in notes 26–27 are consistent with this conjecture; a clean closed form for the kernel of the combined Part II + Part III system is the next target.